

2026-09-14

Easily construct, covert or build Likert items from vectors or data.frame

Job Nmadu 

Federal University of Technology, Minna, Nigeria

This template was last revised on 2026-03-24.

Executive Summary

This proposal outlines a project to develop open-source R package to convert and construct Likert items from vectors or data frames, particularly from non-standard data structures. The proposal aims to bridge the gap where fragmented, manual workarounds that disrupt reproducible data workflows and introduce risks of manipulation errors are utilized for analyzing Likert-type data. The project will be executed over a nine-month timeline with a total proposed budget of USD 8,500. The timeline is divided into an initial two-month inception phase to design the architecture, a four-month coding and testing phase to release the core engine, a two-month preliminary release phase for initial CRAN submission and documentation setup, and a final month for project closure. The budget directly funds development labor at a rate of USD 50 per hour, documentation hosting, and a sensitization workshop for faculty and students at the Federal University of Technology, Minna. Upon completion, the package and its continuous updates will be maintained publicly in GitHub and CRAN repositories to optimize workflow efficiency for social and behavioral science researchers.

Signatories

Project team

Professor Job Nmadu is leading the task and would deliver on the targets.

Contributors

Dr. Halima Sallawu of Federal University of Technology, Minna, Nigeria is in my lab and has participated actively in several data collection activities by which the need for this project is borne.

Consulted

The Problem

The CRAN packages for Likert-type data analysis workflow within the R statistical computing ecosystem, viz, Mollazehi (2025); Winzar (2026) and Bryer and Speerschneider (2025), have not contemplated a situation whereby the items may be transliterated to other formats for more robust analysis and interpretation. They effectively facilitate standard Likert scale workflows, but operate under the assumption that input data is structured in the conventional, clean ordinal format. Consequently, the existing software landscape lacks the architectural framework required to handle data format transliteration.

In empirical research, investigators frequently encounter non-standard data structures that require transformation to achieve robust analysis and deep interpretation. For instance, researchers often face the need to convert Likert text responses into ordinal Likert scales and vice versa, or synthesize valid Likert-type scales from non-Likert source data. Because current R tools cannot execute these complex transformations natively, research teams are forced to rely on disjointed, manual workarounds and ad-hoc scripting. This fragmentation severely disrupts reproducible data workflows, introduces significant risks of data entry and manipulation errors, and results in substantial losses in operational man-hours. This proposed project addresses this computational gap by proposing a novel, open-source R package framework. By proposing a bi-directional transliteration and dynamic format conversion facilities, the software will eliminate manual data manipulation, preserve data integrity, and significantly optimize workflow efficiency for researchers handling complex, non-standard behavioral and social science data.

The proposal

This project aims to provide algorithms around the following concepts:

- binary construction
- likert construction
- likert conversion
- vector description
- weight construction

Overview

The proposal is to create and submit an R package dealing with likert-type data conversions to different formats.

Detail

Minimum Viable Product

The minimum goals is to have a lightweight R package hosted on GitHub that successfully executes bi-directional transliteration between raw character vectors and structured ordinal Likert scales, focusing on likert construction and likert conversion modules.

Architecture

The package will be built within the R statistical computing ecosystem with minimal external dependencies. It will feature an optimized core chunking engine designed to pass integration tests with local memory backends. The internal logic will revolve around transliteration and dynamic conversion algorithms around the main concepts and function of this project.

Assumptions

There is high expectation that the project would be delivered with no particular risk associated with all timelines, except emergencies or natural disasters, like COVID-19 happens.

External dependencies

The proposed package would have minimum dependencies but would rely of **tidyverse** established routines and workflows.

Project plan

Start-up phase

Technical delivery

Functions and codes in GitHub and CRAN repositories as well as continuous development and integration on feedback, stakeholder concerns and updates occasioned by change of roles by R Core teams.

Other aspects

Announcement posts and delivery blogs shall be used for continuous feedback while social media posts will be used for stakeholder engagement.

UseR! conferences and ISC calls are also channels for publicity.

Budget & funding plan

Milestone	Expected Completion period	Deliverables / Tangible Work Products
Inception and take-off	Two months	Complete architecture design and initial abstract class definitions in the GitHub repository.
Coding and testing	Four months	Release the optimized core chunking engine; pass integration tests with local memory backends. Conduct a sensitization workshop for faculty and students.
Preliminary release and initial CRAN submission	Two months	Comprehensive pkgdown documentation, website hosting, initial submission of version 1.0.0 to CRAN, and host a community webinar.
Final release and closure	One month	Final CRAN submission and project closure

In line with the above timelines, the budget proposed for this project is USD8500 appropriated as follows:

Inception and take-off (USD1500): Funds 30 hours of core development work at a rate of USD50/hour. This covers GitHub setup and integration with local machine codes and designing the user-facing API wrappers

Coding and testing (USD5000): Funds 100 hours of setting up functions and algorithms, testing them for consistency and reproducibility and integrating with GitHub repositories. Setting up documentation and website. Initial submission to CRAN.

Sensitization workshop(USD1000): Sensitization workshop and stakeholder engagement for faculty and students at Federal University of Technology, Minna. Publicity over social media handles.

Final release and closure (USD1000): Funds 20 hours for incorporating views from stakeholders into the the package, responding to CRAN reviews and comments, cleaning up the package and final release to CRAN.

Success

Definition of done

Measuring success

Future work

Bryer, Jason, and Kimberly Speerschneider. 2025. *Likert: Analysis and Visualization Likert Items*. <https://doi.org/10.32614/CRAN.package.likert>.

Mollazehi, Mohammad. 2025. *LikertEZ: Easy Analysis and Visualization of Likert Scale Data*. <https://doi.org/10.32614/CRAN.package.LikertEZ>.

Winzar, Hume. 2026. *LikertMakeR: Synthesise and Correlate Likert Scale and Rating-Scale Data Based on Summary Statistics*. <https://doi.org/10.32614/CRAN.package.LikertMakeR>.